

AJAY KUMAR, Ph.D.

Assistant Professor of Physics

Department of Physics, Institute of Science, Banaras Hindu University, Varanasi, UP - 221005, India

Email: ajay.phy@bhu.ac.in | Phone: +91-88-265-265-47

Profiles: [INSPIRE-HEP Profile](#) | [BHU Faculty Profile](#)

PROFESSIONAL SUMMARY

Assistant Professor of Physics at Banaras Hindu University with extensive expertise in experimental high-energy physics, specializing in the CMS Experiment at CERN, the Electron-Ion Collider (EIC), and relativistic heavy-ion collision modeling. Past foundational research includes Higgs boson searches and silicon sensor tracking infrastructure development. Current research programs focus on frontier detector technologies (GEM and LGAD simulation systems) alongside detailed nucleon structure frameworks, supported by a strong publication track record and deep training in particle and computational physics pedagogy.

ACADEMIC APPOINTMENTS

- **Assistant Professor**

Department of Physics, Institute of Science, Banaras Hindu University, Varanasi, India 07/2020 – Present
Instruct graduate (PG) and undergraduate (UG) core courses in Particle Physics, Mathematical Physics, Computational Physics, Quantum Mechanics, and Electronics. Direct advanced practical laboratory workflows across Nuclear Physics, Optics, and Electronics instrumentation. Serve as institutional academic advisor, manage examination execution, and contribute directly to departmental curriculum architecture.

- **Assistant Professor**

Department of Physics, Sri Aurobindo College, University of Delhi, India 08/2016 – 07/2020
Delivered structural lectures in Elements of Modern Physics, Solid State Physics, Atomic Physics, and Computational Physics methods. Supervised core physics practicals across Thermal Physics, Mechanics, and Electricity/Magnetism frameworks.

- **Research Associate**

Department of Physics & Astrophysics, University of Delhi, India 07/2016 – 07/2020
Collaborated across multiple active projects including setting up the Silicon Sensor Qualification Center (SQC), managing Alibava analytical data pipelines, and implementing multivariate search matrices for heavy vector-like top quark partners.

RESEARCH SUPERVISION

Ph.D. Scholars Supervising

- **Rajiv Gupta**

Research Vector: Micro-pattern gaseous detector optimization, finite-element structural simulation via Garfield++/ANSYS, and electrical performance characterization profiles of LGAD Si sensors for upcoming high-luminosity tracking upgrades.

- **Sunidhi Saxena**

Research Vector: Heavy hadron production phenomenology and multi-dimensional anisotropic flow signature mapping inside hot Quark-Gluon Plasma (QGP) matrices utilizing computational frameworks (HYDJET++).

EDUCATION

- **Ph.D. in Physics (Experimental High Energy Physics)**

University of Delhi, New Delhi, India

10/2010 – 11/2016

Thesis: “Search for the SM Higgs Boson in $H \rightarrow WW \rightarrow l\nu jj$ and Probe of WW Production in Vector Boson Fusion Topology in the CMS Experiment at LHC”

Supervisor: Prof. Kirti Ranjan

Date of Award: 16 November 2016

- **M.Sc. in Physics**
University of Delhi, New Delhi, India 07/2008 – 06/2010
Classification: First Division (66.2%)
- **B.Sc. in Physics, Chemistry, Mathematics**
Acharya Narendra Dev College, University of Delhi, India 07/2005 – 06/2008
Classification: First Division (74.44%) *Awarded Certificate of Merit for Topping College*

PRE-DOCTORAL RESEARCH FELLOWSHIPS & INTERNATIONAL VISITS

- **Pre-Doctoral Research Fellow** | Northwestern University, Evanston, USA 03/2016 – 05/2016
Focused on the CMS High-Granularity Calorimeter (HGCAL) Phase-2 upgrades. Developed Data Acquisition Systems (DAQ) and online Data Quality Monitoring (DQM) mechanisms for the test-beam setup at Fermilab, utilizing the OTS-DAQ system in collaboration with Fermilab engineers. Participated actively in test-beam data runs.
- **Visiting Scientist** | LHC Physics Center (LPC), Fermilab, Batavia, USA 10/2015 – 02/2016
Designed multivariate configuration matrices (Boosted Decision Tree models) for the search for heavy vector-like top quark partners in single-lepton final states alongside Fermilab and Brown University teams.
- **Research Intern** | LHC Physics Center (LPC), Fermilab, Batavia, USA 08/2013 – 08/2014
Conducted extensive analysis workflows regarding Standard Model Higgs boson semileptonically decaying channels, diboson cross-section validations, and vector boson scattering topologies.

DETAILED RESEARCH EXPERIENCE & TECHNICAL EXPERTISE

Current Structural Projects (Banaras Hindu University)

- **Simulation and Optimization of GEM Detectors (2023 – Present):** Applying finite-element modeling software (ANSYS) matched with microscopic gas simulation engines (Garfield++) to structurally evaluate electric field lines, gas gain stabilization, and avalanche characteristics to optimize Gas Electron Multiplier geometries.
- **Simulation of Low-Gain Avalanche Detectors (LGADs) (2023 – Present):** Developing comprehensive Technology Computer-Aided Design (TCAD) semiconductor profiles to optimize timing resolution, charge collection mechanisms, and radiation hardness thresholds for the CMS Phase-2 Endcap Timing Layer upgrades.
- **Heavy Hadron Dynamics in Relativistic Collisions (2024 – Present):** Studying heavy hadron tracking production parameters inside Au+Au collision media at $\sqrt{s_{NN}} = 200$ GeV using high-performance Monte Carlo event models (HYDJET++) to extract transport vectors of the Quark-Gluon Plasma.
- **Deeply Virtual Compton Scattering (DVCS) for EIC Physics (2022 – Present):** Modeling structural generalized parton distributions (GPDs) for nucleon structure mappings at the upcoming Electron-Ion Collider. Writing object-oriented analysis software in Python and ROOT to specify the tracking configuration criteria for the ATHENA detector program.

Ph.D. Analysis & Operations (CMS Experiment, CERN)

- **Search for the Standard Model Higgs Boson ($H \rightarrow WW \rightarrow \ell\nu jj$):** Led primary data analysis on semileptonic Higgs decay paths, managing signal extraction optimization, object identification, and systematic error tracking. Extracted background normalizations via unbinned maximum likelihood fits over dijet invariant mass sidebands (m_{jj}), culminating in a landmark high-mass Higgs bounds publication in *JHEP*.
- **Vector Boson Scattering & Diboson Cross-Sections:** Evaluated $WW + WZ$ cross-section tracking matrices in $p-p$ encounters at $\sqrt{s} = 8$ TeV. Formulated custom VBS forward-jet tagging routines to safely isolate rare electroweak processes from overwhelming QCD multijet backgrounds in semileptonic final states.
- **Alignment Parameter Error (APE) Main Contact Expert (2011 – 2015):** Appointed as the primary contact expert for tracking matrix alignment parameter error estimation within the global CMS tracker group. Managed residual minimization routines to calibrate spatial tracker errors, providing optimal track resolution during active data takes.

- **Tracker Operations & Offline DQM Shifts:** Performed offline Data Quality Monitoring (DQM) and track validation shifts during active LHC physics runs. Calibrated b-tagging scale factor constants and monitored cosmic ray data profiles.
- **Calorimeter Response & Single Particle Studies:** Measured single particle isolation energy response trends across the CMS Electromagnetic (ECAL) and Hadronic (HCAL) Calorimeters at 8 TeV. Code outputs directly contributed to Geant4 simulation adjustments and jet energy scale refinements (*CMS-DN-14-030*).
- **HLT Laser Correction Validation (2015 Data-Taking):** Served as the central validation contact for High-Level Trigger (HLT) laser corrections, monitoring transparency changes in lead-tungstate crystal matrix arrays under high radiation loads via specialized graphical monitoring interfaces.

Hardware Lab Infrastructure (University of Delhi)

- **Sensor Qualification Center (SQC) Development (2016 – 2020):** Co-led the physical assembly and development of the automated Silicon Sensor Characterization hardware lab at Delhi University for future high-energy experiments. Authored automated IV and CV extraction scripts in Python and Lua, interfacing Keithley instruments with semi-automated MPI probe stations. System received formal QA/QC certification from CMS tracker experts during the 3rd global SQC workshop.
- **Alibava Readout Setup:** Formulated data parsing routines for silicon multi-strip sensors read out via Alibava DAQ chips, enabling charge collection modeling for irradiated tracking elements.

SELECTED PUBLICATIONS

Journal Articles

- Sunidhi Saxena, Rajiv Gupta, G. Devi, and **Ajay Kumar** (2026). Anisotropic flow of heavy hadrons in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV using HYDJET++ framework. *The European Physical Journal Plus*, Accepted/In Press.
- Sunidhi Saxena, Rajiv Gupta, G. Devi, and **Ajay Kumar** (2026). Heavy hadron production in Au + Au collisions at a center-of-mass energy of 200 GeV. *Physical Review C*, 113, 014905.
- Rajiv Gupta, G. Devi, Sunidhi Saxena, A. Singh, and **Ajay Kumar** (2025). Simulation-based performance comparison of varied pitch sizes GEM detectors. *Nuclear Instruments and Methods in Physics Research Section A*, 1082, 171046.
- J. Adam et al. [ATHENA Collaboration] (2022). ATHENA detector proposal — a totally hermetic electron nucleus apparatus proposed for IP6 at the Electron-Ion Collider. *Journal of Instrumentation*, 17, P10019.
- G. Jain, C. Jain, **A. Kumar**, et al. (2019). Radiation hardness investigation of thin and low resistivity bulk Si detectors. *Nuclear Instruments and Methods in Physics Research Section A*, 935, 559–561.
- CMS Collaboration (2017). Search for anomalous couplings in boosted WW/WZ $\rightarrow$ lvqq production in proton-proton collisions at $\sqrt{s} = 8$ TeV. *Physics Letters B*, 772, 21–42.
- CMS Collaboration (2015). Search for a Higgs boson in the mass range from 145 to 1000 GeV decaying to a pair of W or Z bosons. *Journal of High Energy Physics*, 10, 144.
- CMS Collaboration (2014). Search for WW γ and WZ γ production and constraints on anomalous quartic gauge couplings in pp collisions at $\sqrt{s} = 8$ TeV. *Physical Review D*, 90, 032008.
- CMS Collaboration (2014). Alignment of the CMS tracker with LHC and cosmic ray data. *Journal of Instrumentation*, 9, P06009.

Conference Proceedings

- Rajiv Gupta, **A. Kumar**, et al. (2024). Investigating the effects of acceptor removal mechanism and impact ionization on proton irradiated 300 μ m thick LGAD. *Proc. DAE Symposium on Nuclear Physics*, 68, 1070.
- G. Jain, **A. Kumar**, et al. (2019). Development of an automated and programmable characterization system for silicon multi-strip sensors. *Nuclear Instruments and Methods in Physics Research Section A*, 936, 663–665.
- C. Jain, **A. Kumar**, et al. (2019). Performance of silicon sensor quality control centre developed at the University of Delhi. *IEEE Proceedings*, 978-1-7281-4164-0.

- **A. Kumar** (2014). Measurement of Electroweak Vector Boson Pair Production in p-p Collision with the CMS Detector at LHC. *Proc. Second Annual LHCP*, CMS-CR-2014/193.

TECHNICAL & COMPUTATIONAL SKILLS

- **Languages:** C++, Python (NumPy, Pandas, Matplotlib, SciKit-Learn), Lua, Shell Scripting
- **High-Energy Physics Analysis Tools:** ROOT, CMSSW, RooFit, TMVA, Higgs Combine Tool
- **Simulation Architectures:** ANSYS, Garfield++, TCAD (LGAD profiling), Geant4, PYTHIA8, MadGraph, HYDJET++
- **Frameworks & Instrumentation Software:** Git, LaTeX, Keithley Instrument Control, MPI Probe Control Interfaces

AWARDS, HONORS & ACHIEVEMENTS

- **Junior Research Fellow (Non-NET JRF):** University Grants Commission (UGC), India 2010 – 2013
- **National Eligibility Test (NET) Qualification:** UGC-CSIR Joint Examination, India 2010
- **Certificate of Academic Merit:** Awarded for Topping College in B.Sc., Acharya Narendra Dev College

SELECTED CONFERENCES & PRESENTATIONS

- **Poster Presentation:** “*Simulation study of GEM detectors*”, XXVI DAE-BRNS High Energy Physics Symposium, India 2024
- **Oral Presentation:** “*TCAD simulation of proton-irradiated LGADs*”, XXVI DAE-BRNS High Energy Physics Symposium, India 2024
- **Invited Expert & Participant:** Third Global Sensor Quality Control (SQC) Workshop, University of Delhi, India 01/2019
- **Participant Focus:** AMDFA-2018, Chandigarh University, India 05/2018